

# Payout Mechanisms as Operators on Attention Distributions: Identities, Invariances, and Tail Action

Tonify Research

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## 1 Setup and the generative model of sim4

Artists  $i \in N$ , listeners  $u \in U$  ( $|U| = U$ ), integer play matrix  $p = (p_{iu}) \in \mathbb{Z}_+^{N \times U}$  over one period. Row/column sums:  $P_i = \sum_u p_{iu}$ ,  $P_u = \sum_i p_{iu}$ ,  $T = \sum_u P_u$ , mean intensity  $\bar{P} = T/U$ . Equal wallets  $W > 0$ ; pool  $R = UW$ .

**Assumption 1** (A1, load-bearing). *Every paying subscriber listens:  $P_u \geq 1$ . Silent payers are excluded from both pools; otherwise budget balance below fails by exactly  $W \cdot U_{\text{sil}}$  and all ratios acquire the factor  $U_{\text{act}}/U_{\text{tot}}$ .*

**Mechanisms.** Pro-rata  $\text{PR}_i = UW P_i/T$ ; user-centric  $\text{UC}_i = W \sum_u p_{iu}/P_u$ ; direct  $D_i = \tau \sum_{f=1}^{S_i} \sum_{t=1}^k G_{ft}$  with  $S_i | A_i \sim \text{Bin}(A_i, \sigma)$ , audience  $A_i$ , i.i.d. gifts  $G \geq 0$ ,  $\mathbb{E}G = \bar{g}$ , all moments finite; artist share  $\tau$ . A label contract acts multiplicatively after the rule:  $\rho \cdot F_i(p)$ .

**Generative model (sim4).** Artist popularity: piecewise  $s_i \sim (\text{lognormal body} / \text{log-bridge} / \text{Pareto}(\alpha=1.4) \text{ tail})$ , calibrated to anchors  $T_1$ : 87% of artists below  $10^3$  streams/yr,  $T_2$ : 2.6% above 225,734,  $T_3$ : top 0.28% holding 40–55% of streams (gate band; the seed-42 matrix realizes 40.3%, sim1’s larger world 44.5%); target audiences  $\ell_i = s_i/\text{PL}$ ,  $\text{PL} = 21.21$  plays per pair (LFM-1b). Playlist sizes  $K_u \sim \text{LogNormal}(\ln 12, 1.0)$  (assumed; an MVP-measured dial). Artist selection: exact weighted sampling without replacement,  $\mathbb{P}(i \in S_u) \propto \ell_i$  (Gumbel top- $K_u$ ), i.e. preferential attachment. Pair intensity, three named regimes on one graph:

(a) ergodic: integer  $p_{iu}$ ,  $P_u \equiv 2048$ ; (b)  $p_{iu} = \lceil \text{LN}(\ln 5.16, 1.676) \rceil$  rescaled to  $s_i$ ; (c)  $p_{iu} \propto \text{LN} \cdot (A_i/\bar{A}_g)^\gamma$ .

Regime (a) has  $P_u \equiv 2048$  exactly (a power of two: every  $p_{iu}/P_u$  is a dyadic rational, so the control identity is exact in floating point) and is the control (Corollary (a));  $\gamma$  is the intensity–size coupling dial of §4.

## 2 The ratio identity

**Theorem 1** (Identity). *For any artist  $i$  with  $P_i > 0$ , under equal wallets,*

$$\frac{\text{UC}_i}{\text{PR}_i} = \mathbb{E}_{w^i} \left[ \frac{\bar{P}}{P_u} \right], \quad w_u^i = \frac{p_{iu}}{P_i}.$$

*Proof.*  $\text{UC}_i/\text{PR}_i = (W \sum_u p_{iu}/P_u) / (UW P_i/T) = (T/U) \sum_u (p_{iu}/P_i)(1/P_u) = \bar{P} \mathbb{E}_{w^i}[1/P_u]$ .  $\square$

With  $m_i = \mathbb{E}_{w^i}[P_u]$  (arithmetic) and  $H_i = 1/\mathbb{E}_{w^i}[1/P_u]$  (harmonic mean of listener intensity under play weights),

$$\frac{\text{UC}_i}{\text{PR}_i} = \frac{\bar{P}}{H_i} = \underbrace{\frac{\bar{P}}{m_i}}_{\text{level}} \cdot \underbrace{\frac{m_i}{H_i}}_{\text{dispersion}} \geq 1,$$

the second factor  $\geq 1$  by AM–HM, with equality iff  $P_u$  is constant on the support of  $w^i$ .

**Corollary (a)** (Coincidence; the naive converse is false).  $P_u \equiv \bar{P}$  implies  $\text{UC} = \text{PR}$  componentwise. The converse fails: with one artist and two listeners,  $p = (10, 20)$ , intensities differ yet  $\text{UC}_1 = \text{PR}_1 = 2W$ . Exact criterion:  $\text{UC}_i = \text{PR}_i$  for all  $i$  iff the vector  $c_u = 1/P_u - U/T$  lies in  $\ker p$ .

**Corollary (b)** (Zero sum).  $\sum_i \text{UC}_i = \sum_i \text{PR}_i = UW$ : user-centric is a pure redistribution of the same pool.

**Corollary (c)** (Crossover).  $\text{UC}_i > \text{PR}_i$  iff  $H_i < \bar{P}$ . On type populations this reduces to the condition of Alaei, Makhdoumi, Malekian and Pekeč (Mgmt. Sci. 2022, Props. 2, 8): artist  $j$  weakly prefers pro-rata iff  $\sum_i q_i \pi_{ij} (\lambda_i - \bar{\lambda}) \geq 0$ .

**Corollary (d)** (Dispersion premium). Fix  $m_i = \bar{P}$ . Then  $\text{UC}_i/\text{PR}_i = m_i/H_i \geq 1$ , with equality iff audience intensity is degenerate: at the same mean level, within-audience dispersion strictly raises user-centric income.

### 3 Pass-through invariance

**Lemma 2** (Two-part invariance). (2a) For any pool operator  $F$  applied at the artist level, with the multiplicative contract strictly downstream and on the domain  $F_i(p) > 0$ :  $\rho F_i/F_i = \rho$  the signed/independent income gap is invariant to the rule (a nonlinear  $F$  applied to a licensor’s catalogue breaks this). (2b) The minimum-viable-audience gap satisfies  $\text{MVA}_s/\text{MVA}_l = 1/\rho$  iff  $F^{-1}(V/\rho) = F^{-1}(V)/\rho$ ; linearity of  $F$ -income in audience is sufficient but not necessary (a strictly increasing log-periodic  $F$  is nonlinear yet yields  $1/\rho$  for all  $V$ ). A Deezer-style  $\times 2$  boost above a threshold yields  $1/(2\rho)$  instead (counterexample in companion code) while the income ratio stays  $\rho$  pointwise.

Breaking channels: endogenous  $\rho(F)$ ; additive components; payout thresholds (accumulation delays  $\times 1/\rho$  in time); behavioral endogeneity of  $p$ ; component/market-heterogeneous  $\rho_{\text{eff}} = \sum_m \rho_m w_{im}(F)$ .

### 4 Tail action and the atom at zero

Let  $\mathbb{P}(A > x) = x^{-\alpha} L(x)$ . PR is linear: index preserved. UC:  $\text{UC}_i = W P_i \mathbb{E}_{w^i}[1/P_u]$  with bounded multiplier; under tail-conditional concentration of the multiplier the index is preserved, and heavier tails are impossible for any dependence. Direct: binomial thinning plus a light-tailed compound sum preserve  $\alpha$  (Jessen–Mikosch 2006, Lem. 3.7; Faÿ et al. 2006; Denisov–Foss–Korshunov 2010; Breiman 1965 enters only through the multiplicative contract layer), while  $\mathbb{P}(D_i = 0 | A_i) \geq (1 - \sigma)^{A_i} \geq e^{-1}(1 - \sigma)$  for  $A_i \leq 1/\sigma$ : an *atom at zero* on small audiences. With zero mass  $q$ ,  $G_{\text{direct}} = q + (1 - q)G^+$ : the mechanism relocates inequality from the intensive to the extensive margin; no mechanism softens the upper tail.

## 5 Dominance blockade

**Theorem 4** (Full blockade). *Fix  $A$  with  $(1 - \sigma)^A > 0$  and pooled income  $X_A = cA > 0$  (deterministic given  $A$ );  $Y_A = D_A$  as above. Then: (i)  $Y_A$  never first-order dominates  $X_A$ ; (ii)  $Y_A$  never second-order dominates  $X_A$ , for any parameters; (iii)  $X_A \succeq_2 Y_A$  iff  $\mathbb{E}Y_A \leq X_A$ , i.e. iff  $k \leq d^* = \text{PL} \cdot \text{rate} / (\sigma \bar{g} \tau)$ ; (iv) for  $k > d^*$  the pair is incomparable in both orders.*

*Proof.* (i) For  $x \in (0, X_A)$ :  $\mathbb{P}(X_A > x) = 1$  while  $\mathbb{P}(Y_A > x) \leq 1 - (1 - \sigma)^A < 1$ . (ii) SSD against degenerate  $X_A$  requires  $\int_0^t F_Y \leq \max(0, t - X_A)$ ; at  $t = X_A$  the left side is  $\geq (1 - \sigma)^A X_A > 0$ . (iii)  $(\Leftarrow)$   $\int_0^t F_Y = t - \mathbb{E} \min(Y, t) \geq t - \mathbb{E}Y \geq t - X_A$  and  $\geq 0$ .  $(\Rightarrow)$   $t \rightarrow \infty$  gives  $\mathbb{E}Y \leq X_A$ . (iv) From (i)–(iii).  $\square$

*Remark.* The correct comparison is mean versus risk: for  $k > d^*$  direct pays an expectation premium priced by small-audience lottery risk;  $\mathbb{P}(Y_A < X_A) \leq \inf_{t>0} e^{tX_A} (1 - \sigma + \sigma \varphi(t)^k)^A$  with  $\varphi(t) = \mathbb{E}e^{-t\tau G} < 1$  exponential decay in  $A$ . In the companion sim1 calibration at  $A=30$ ,  $k=4$ : mean advantage  $\times 4$ , yet  $\mathbb{P}(Y < X) \approx 0.60$  and on the full 2M-sample run  $\mathbb{P}(Y < X) - \mathbb{P}(Y = 0) = 0$  to machine precision: losing to the pool is exactly the zero outcome.